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High-yield factors in the new soybean cultivar 'Soramizuki' derived from a US parent cultivar

Ryo Yamazaki, Yohei Nanjo, Masayasu Saruta, Shin Kato, Kaori Hirata, Kenta Nakashima, Ryoma Takeshima , Koji Takahashi and Emiko Aoki

Institute of Crop Science, National Agriculture and Food Research Organization (NARO), Tsukuba, Ibaraki, Japan

ABSTRACT

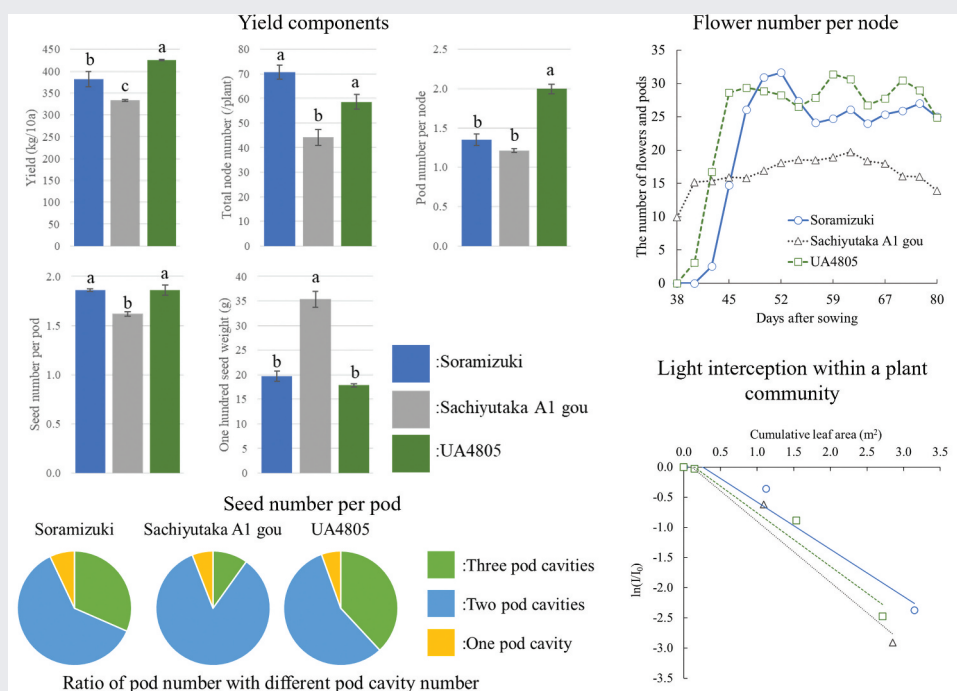
A new soybean cultivar, 'Soramizuki,' was developed by crossing 'Fukuyutaka A1 gou' with 'UA4805' to integrate the high-yield characteristics of a US cultivar into a Japanese cultivar. While the high yield and yield stability of 'Soramizuki' have been confirmed, the specific high-yielding factors were previously unknown. This study compared the yield and yield components of 'Soramizuki,' 'UA4805,' and 'Sachiyutaka A1 gou' in experimental fields during 2021 and 2023. In these two-year experiments, 'Soramizuki' demonstrated a significantly higher yield than 'Sachiyutaka A1 gou.' Despite a significant yield decrease in 'UA4805' in 2023 due to stink bug damage, it showed the highest yield among the three cultivars in 2021. Both 'Soramizuki' and 'UA4805' exhibited more total nodes, pods per node, and seeds per pod compared to 'Sachiyutaka A1 gou.' These differences were attributed to a higher node increase rate, a larger number of blooms per node, and a higher ratio of pods with three seed cavities. Additionally, 'Soramizuki' and 'UA4805' had greater dry matter weight per plant than 'Sachiyutaka A1 gou,' though there were no variations in the harvest index. The light interception ratio per leaf area in the upper canopy was lower in both 'Soramizuki' and 'UA4805.' These results suggest that the high-yielding traits of 'Soramizuki' are derived from the US cultivar 'UA4805.'

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High-yielding traits of 'Soramizuki' suggested to be derived from the US parent cultivar.

CONTACT Ryo Yamazaki  yamazakiroyo@affrc.go.jp

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Introduction

Soybean (*Glycine max* (L.) Merrill) is a crucial crop for traditional Japanese foods such as tofu, soy sauce, natto, and miso. However, Japan's self-sufficiency rate for food-grade soybeans was only 23% in 2022 (MAFF, 2023). Thus, it is essential to increase soybean yields in Japan by breeding high-yielding cultivars.

While soybean yields in the US have been rising, yields in Japan have remained stagnant (FAO, 2023; Katsura et al., 2009). Approximately 50% of the yield increase in the US over the past 80 years has been attributed to breeding advancements (Boehm et al., 2019). In comparative studies, US cultivars demonstrated higher yield capacities than Japanese cultivars when grown in Japan (Kawasaki et al., 2016; Kumagai et al., 2022; Matsuo et al., 2016). The higher yield of US cultivars has been linked to greater total aboveground dry matter (Kawasaki et al., 2016) and a higher number of pods m^{-2} (Matsuo et al., 2016). Additionally, Kumagai et al. (2022) suggested that smaller seed size in US cultivars might contribute to higher yields due to a trade-off between seed size and yield. Tanaka et al. (2010) proposed that US cultivars have a greater photosynthetic capacity because of higher leaf stomatal conductance.

A new soybean cultivar, 'Soramizuki,' was developed by crossing 'Fukuyutaka A1 gou' with 'UA4805' to incorporate high-yielding genetic factors from US cultivars into Japanese varieties. This breeding effort was conducted by the Institute of Crop Science, National Agriculture and Food Research Organization (NARO) (Kato et al., 2023). 'UA4805' is a determinate US cultivar that has shown high yields in Japanese experimental fields (Kawasaki et al., 2016; Matsuo et al., 2016). 'Fukuyutaka A1 gou' is a cultivar with a pod dehiscence resistance gene (*pdh1*), developed from 'Fukuyutaka,' which is the most widely cultivated Japanese cultivar (Funatsuki et al., 2006; Hajika et al., 2019).

'Soramizuki' is a determinate cultivar with pod dehiscence resistance and the same maturity as the Japanese cultivar 'Sachiyutaka A1 gou' (Hajika et al., 2016). Although its seed size is smaller than that of standard Japanese cultivars, its suitability for tofu processing is comparable to 'Fukuyutaka' (Kato et al., 2023). Kato et al. (2023) demonstrated the high yield and seed quality of 'Soramizuki.' However, the specific high-yielding factors of 'Soramizuki' have not been clarified.

The development of future cultivars that exceed the yield potential of 'Soramizuki' using high-yielding US varieties is anticipated. Therefore, identifying the phenotypes related to the high-yielding factors of 'Soramizuki' is crucial for improving breeding efficiency. High-yielding phenotypes are instrumental in identifying

high-yielding genetic factors. This study focuses on yield components, dry matter production, harvest index, and light interception characteristics to compare these phenotypes among 'Soramizuki,' a Japanese cultivar, and the parent US cultivar.

Materials and methods

Plant materials

'Soramizuki' (Kato et al., 2023), 'Sachiyutaka A1 gou' (Hajika et al., 2016), and 'UA4805' (Chen et al., 2006) were used in this study. 'Soramizuki' is a determinate cultivar with pod dehiscence resistance. 'Sachiyutaka A1 gou' is a determinate cultivar with pod dehiscence resistance and leading cultivar in Japan with an almost identical maturity period to 'Soramizuki'. 'UA4805' is a determinate high-yielding US soybean cultivar with pod dehiscence resistance and one of the parents of 'Soramizuki'.

Experimental conditions and measurements

Experiment 1: yield and yield components in the field

Experiment 1 was conducted at the experimental fields of the NARO Institute of Crop Science, Ibaraki, Japan (36°0'N, 140°1'E; Elevation: 11.4 m in 2021 and 11.1 m in 2023; Soil: Andosol-Brown). The experiment was repeated over 2 years (2021 and 2023) in a randomized block design with three replications. Sowing dates were June 11 in 2021 and June 21 in 2023. Seeds were sown after adding fertilizer (3 g m^{-2} N, 10 g m^{-2} P_2O_5 , and 10 g m^{-2} K_2O) to the soil. Each plot measured 2.1 m $\times$ 1.5 m in 2021 and 5.6 m $\times$ 4.0 m in 2023, with a plant density of 11.0 plants m^{-2} (0.7 m $\times$ 0.13 m). Insecticides were applied regularly from July to September in both years.

Meteorological data (daily temperature, solar radiation, and precipitation) recorded at the experimental site by NARO were utilized. The dates for the R1 (beginning bloom), R5 (beginning seed filling), and R8 (full maturity) stages for each plot were determined using the method described by Fehr and Caviness (1977). On the day each plot reached the R8 stage, the lodging angle to the vertical line was evaluated using the method described in Kato et al. (2023), where six scores (0–5) were assigned to each plot. After the R8 stage, 12 plants (excluding plot borders) were randomly selected for the analysis of yield and yield components. Seed weight, seed number, pod number, pod cavity number, total node number, pod dry weight, stem dry weight, and main stem length were measured in each plot. Dry weights were recorded after drying the samples at 80°C for 3 days. Seed moisture content was analyzed using a NIR grain analyzer

(Inframatic 9500, Perten) to convert seed weight to an equivalent at 15% moisture.

Two plants in 2021 and six plants in 2023, randomly selected excluding borders, were sampled on 6 September 2021, and 7 September 2023 (the R5 stage). The leaf area and leaf number at the R5 stage were measured using an area meter LI-3100C (LI-COR, Lincoln, NE, US).

In 2023, the sampled plants at the R5 stage were refrigerated at 5°C before wilting. The main stem of sample was fixed vertically in the research room and the sampled plant was surrounded by metal frame at every 20 cm from the ground. Using the frame as a guide, leaves of the sampled plant cut from the uppermost layer and separated into 20 cm layers, and the leaf area of each layer was measured using an area meter LI-3100C. Photosynthetically active radiation (PAR) at every 20 cm from the ground in the canopy at the R5 stage was measured in 2023 using an LI-191 R Line Quantum Sensor (LI-COR, Lincoln, NE, US). Measurements were conducted between 10:00 and 13:00. Using two line quantum sensors, one placed inside the canopy and another outside, PAR was simultaneously recorded. For each plot, the line quantum sensor in the canopy was placed across two ridges, excluding borders, and recorded PAR at every 20 cm until the sensor exceeded the canopy height. The canopy height at the center of the line quantum sensor was also recorded.

Experiment 2: cultivar differences in the number of flowers and pods after the R1 stage

A pot experiment was conducted in a netted house at the NARO Institute of Crop Science, Ibaraki, Japan (36°1'N, 140°6'E; Elevation: 22.7 m) in 2023. The netted house maintained a temperature nearly identical to the outside environment because only the roof was covered with a fluoropolymer film (F-CLEAN GR80, AGC Chemicals). Polyethylene pots (28 cm diameter and 26 cm height) were filled with 7 kg of planting soil for crops (Tsuchitaro, SUMITOMO FORESTRY CO., LTD.) and placed on bottom-watering trays. Water was added to the trays to maintain a depth of 4.5 cm above the bottom of the pots. Sowing took place on July 7. In each pot, 18 seeds were sown. Seedlings were thinned to three plants at 6 days after sowing (DAS), and the most moderately growing plant was selected at 13 DAS. A randomized block design with 11 replications (11 pots per cultivar) was used. Plant density (interval between pots) was 4.2 plants m⁻² (0.8 m × 0.3 m).

It was reported that there tended to be more flowers per pod in the upper part of the main stem than

in the lower part of the main stem or in branches (Takahashi et al., 1978). Thus, to align conditions for comparison, the uppermost three nodes in the main stem was compared among the three cultivars. Starting on August 11, approximately corresponding to the R2 stage (full bloom) for each cultivar (Fehr & Caviness, 1977), the number of flowers or pods on the uppermost three nodes of the main stem was recorded every 2 to 3 days. The count included all flowers and pods on the uppermost three nodes that continued growing, including closed flowers after blooming. Thus, an increase in the number of flowers or pods indicated that newly blooming flowers exceeded shedding flowers or pods during the interval between measurements, while a decrease indicated that shedding flowers or pods exceeded newly blooming flowers.

Recording ceased on September 15 when no new flowers were blooming on the uppermost three nodes in any cultivar. After each plant reached the R8 stage, the number of pods, seeds, and pod cavities on the uppermost three nodes were measured.

Statistical analysis

The experiments were conducted using a randomized block design. Statistical significance of differences between years and cultivars was evaluated by Type III analysis of variance (ANOVA) at significance levels of $p < 0.05$ or $p < 0.01$. When ANOVA indicated significant differences, Tukey's tests were used to compare parameters among the three cultivars. All analyses were performed using BellCurve for Excel version 3.21 (Social Survey Research Information Co. Ltd., <https://bellcurve.jp/ex/>).

Results

Meteorological conditions, insect damage, and growth stages

Figure S1 illustrates the differences in temperature, solar radiation, and precipitation between the 2 years of Experiment 1. Temperatures were higher until late September in 2023 compared to 2021 (Fig. S1A). Solar radiation was also higher until mid-September in 2023 than in 2021 (Fig. S1B). Total precipitation was greater in 2021 than in 2023, except for early and late September (Fig. S1C).

In 2021, the proportion of seeds damaged by pod-feeding stink bugs (Fig. S2) ranged from 0.6% to 2.4% (Table S1), with no significant differences in damage rates among the cultivars. In 2023, 'UA4805' showed

a significantly higher proportion of seeds damaged by pod-feeding stink bugs, while 'Soramizuki' and 'Sachiyutaka A1 gou' exhibited damage rates similar to those in 2021 (1.7% and 1.1%, respectively) (Table S1). This indicates that in 2023, only 'UA4805' experienced significant damage from pod-feeding stink bugs.

'Soramizuki' and 'Sachiyutaka A1 gou' exhibited similar growth stage dates in both 2021 and 2023 (Table 1). In 2021, the R5 and R8 stages of 'UA4805' occurred earlier than those of the other two cultivars (Table 1). However, in 2023, the R8 stage of 'UA4805' was later than that of the other two cultivars (Table 1).

Yield and yield components

The yield and yield components of the three cultivars across 2 years are presented in Table 2. The ANOVA results indicated no significant cultivar differences in yield; however, there were statistically significant differences between years and their interaction. In 2021, 'UA4805' had the highest yield, followed by 'Soramizuki' and 'Sachiyutaka A1 gou'. In 2023, the yield of 'UA4805' was lower than in 2021, whereas the yields of 'Soramizuki' and 'Sachiyutaka A1 gou' showed little variation between years. The standard deviation for 'UA4805' in 2023 (105.1) was substantially higher than that of the other cultivars (2.9–37.7) (Table S2). These results correspond to the significant damage 'UA4805' experienced from pod-feeding stink bugs in 2023 (Table S1). The insect damage in 2023 likely

decreased the yield and increased the yield deviation of 'UA4805', contributing to the lack of statistically significant cultivar differences in the ANOVA.

Cultivar differences, year differences, and their interaction for the weight of 100 seeds were statistically significant (Table 2). The 100 seed weight of 'Sachiyutaka A1 gou' was significantly larger than that of 'Soramizuki' and 'UA4805'. 'Soramizuki' had a slightly larger seed weight than 'UA4805', though the difference was not statistically significant. The 100 seed weight of 'UA4805' in 2023 (12.8 g) was significantly lower than in 2021 (17.9 g).

Both 'UA4805' and 'Soramizuki' had a greater total number of nodes than 'Sachiyutaka A1 gou'. These two cultivars also exhibited higher values in the number of nodes on both the main stem and branches compared to 'Sachiyutaka A1 gou'. Additionally, 'UA4805' and 'Soramizuki' had more pods per node and a higher seed number per pod than 'Sachiyutaka A1 gou'. However, the seed number per pod for 'UA4805' in 2023 (1.4 seeds per pod) was lower than in 2021 (1.9 seeds per pod) or 'Sachiyutaka A1 gou' in 2023 (1.6 seeds per pod). 'UA4805' showed a significantly larger pod number per node compared to 'Soramizuki'.

The lodging angle to the vertical line at the R8 stage was less than 1 (less than 19°) in every plot in Experiment 1 (data not shown), although main stem length was significantly larger in 'Soramizuki' and in 'UA4805' than in 'Sachiyutaka A1 gou' (Table 2). Therefore, the effects of lodging on yield and yield components could be disregarded in this study.

Table 1. Growth stage in experiment 1.

Year	Cultivar	Sowing date	R1	R5	R8
2021	Soramizuki	11-Jun.	2-Aug.	31-Aug.	3-Nov.
	Sachiyutaka A1 gou		1-Aug.	30-Aug.	5-Nov.
	UA4805		2-Aug.	28-Aug.	22-Oct.
2023	Soramizuki	21-Jun.	7-Aug.	3-Sep.	3-Nov.
	Sachiyutaka A1 gou		7-Aug.	4-Sep.	2-Nov.
	UA4805		7-Aug.	26-Aug.	9-Nov.

Table 2. Yield and yield components in experiment 1.

Year	Cultivar	Yield (g m ⁻²)	Node number				Pod number (/node)	Seed number (/pod)	100-seed weight (g)	Main stem length (cm)
			Total node number (/plant)	Node number of main stem (/plant)	Node number of branches (/plant)					
2021	Soramizuki	381.8b	70.7a	16.0a	54.7a		1.3b	1.9a	19.7b	59.7a
	SachiyutakaA1gou	333.3c	44.2b	13.3b	30.8b		1.2b	1.6b	35.3a	43.1b
	UA4805	425.9a	58.6a	15.3b	43.4ab		2.0a	1.9a	17.9b	55.4a
ANOVA		**	**	**	**		**	**	**	**
2023	Soramizuki	385.4	67.7	17.9a	49.4		1.7b	1.8	18.0b	74.3a
	SachiyutakaA1gou	313.5	55.4	13.8c	41.6		1.2c	1.6	26.0a	48.5b
	UA4805	245.7	65.4	15.2b	50.2		1.9a	1.4	12.8b	64.8a
ANOVA		ns	ns	**	ns		**	ns	**	**
Two-way	Year	*	ns	**	ns		ns	*	**	**
ANOVA	Cultivar	ns	**	**	**		**	ns	**	**
	Interaction	*	ns	**	*		**	ns	*	ns

**: Significantly different ($p < 0.01$). *: Significantly different ($p < 0.05$). ns: Nonsignificant ($p < 0.05$). Same letters within the same year indicate no significant differences (Tukey's test, $p < 0.05$).

Flower and pod number per node on the uppermost three nodes of the main stem

To identify the cause of cultivar differences in pod number per node (Table 2), we compared the chronological changes in the number of flowers or pods on the uppermost three nodes of the main stem between cultivars from 38 DAS to 80 DAS (Figure 1).

Both 'Soramizuki' and 'UA4805' exhibited a similar pattern, with the number of flowers or pods repeatedly increasing and decreasing three times, reaching 25 by 80 DAS (Figure 1). The maximum number recorded was 32 for 'Soramizuki' and 31 for 'UA4805'. In contrast, 'Sachiyutaka A1 gou' did not show significant fluctuations in the number of flowers or pods, with the count peaking at 20 and decreasing to 14 by 80 DAS (Figure 1). The number of pods at the R8 stage was almost the same for 'Soramizuki' and 'UA4805', whereas 'Sachiyutaka A1 gou' had significantly fewer pods than the other two cultivars (Table 3). The pod

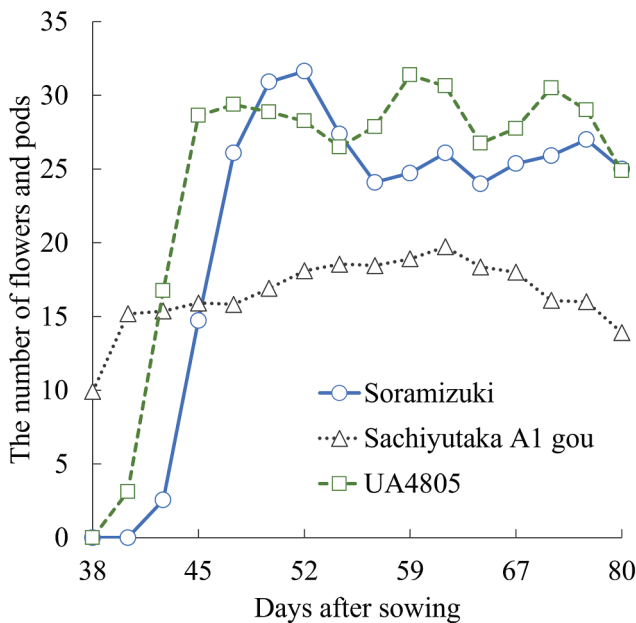


Figure 1. The number of flowers and pods in Experiment 2. The number of flowers or pods represents the total count of flowers and pods that continued growing after blooming on the uppermost three nodes. Each plot represents the average score of 11 plants.

Table 3. Pod number of the uppermost three nodes at the R8 stage in experiment 2.

Cultivar	Pod number
Soramizuki	24.8a
Sachiyutaka A1 gou	14.6b
UA4805	25.2a
ANOVA	**

**Significantly different ($p < 0.01$). Same letters indicate no significant differences (Tukey's test, $p < 0.05$).

counts at 80 DAS (Figure 1) were nearly identical to the counts at the R8 stage for each cultivar (Table 3), indicating that almost no pods were shed from 80 DAS to the R8 stage.

Seed number per pod

The ratio of pods with three seed cavities, two seed cavities, and one seed cavity to the total number of pods is shown in Table 4. 'UA4805' and 'Soramizuki' had significantly higher ratios of three-cavity pods compared to 'Sachiyutaka A1 gou' (Table 4). Conversely, 'UA4805' and 'Soramizuki' exhibited significantly lower ratios of two-cavity pods than 'Sachiyutaka A1 gou'. The ratio of one-cavity pods did not show significant differences among the cultivars in either year. There were no significant interactions between cultivars and years for any pod cavity number.

The seed number per pod cavity did not differ among the cultivars (Table 4), although 'UA4805' had a lower value in 2023 due to damage from pod-feeding stink bugs (Table S1). Thus, the higher ratio of pods with three seed cavities was the primary factor contributing to the larger seed number per pod in 'UA4805' and 'Soramizuki' (Table 2).

Dry matter accumulation

The values related to dry matter accumulation are shown in Table 5. The ANOVA of aboveground dry matter indicated significant differences for the year, cultivar, and their interaction. In 2021, 'UA4805' exhibited the highest aboveground dry matter, followed by 'Soramizuki' and 'Sachiyutaka A1 gou'. In 2023, the aboveground dry matter of 'Soramizuki' was greater than that of 'Sachiyutaka A1 gou', but 'UA4805' had lower aboveground dry matter compared to 'Soramizuki'.

The harvest index, defined as seed weight per total aboveground dry matter, showed no significant differences among the cultivars. Similarly, pod dry weight per seed dry weight did not differ significantly among the cultivars. However, in 2023, 'UA4805' exhibited a lower harvest index (0.33) and a higher pod dry weight per seed dry weight (0.62) compared to the other cultivars (0.46–0.57 and 0.27–0.31, respectively) (Table 5) likely due to stink bug damage (Table S1).

Leaf number, leaf size, and light availability at the R5 stage

The values related to light interception characteristics at the R5 stage are presented in Table 6. There were

Table 4. Pod number with different pod cavity number per 100 pods and seed number per pod cavity in experiment 1.

Year	Cultivar	Pod number with three seed cavities per 100 pods	Pod number with two seed cavities per 100 pods	Pod number with one seed cavity per 100 pods	Seed number per pod cavity
2021	Soramizuki	29.9b	54.2b	7.4	0.87
	Sachiyutaka	9.4c	78.8a	5.9	0.82
	A1 gou				
ANOVA	UA4805	37.4a	50.4b	5.4	0.83
		**	**	ns	ns
2023	Soramizuki	28.4a	59.5b	5.6	0.81
	Sachiyutaka	9.3b	80.2a	5.3	0.82
	A1 gou				
ANOVA	UA4805	33.9a	55.5c	4.8	0.62
		**	**	ns	ns
Two-way ANOVA	Year	ns	**	ns	*
	Cultivar	**	**	ns	*
	Interaction	ns	ns	ns	ns

**: Significantly different ($p < 0.01$). *: Significantly different ($p < 0.05$). ns: Nonsignificant ($p < 0.05$). Same letters within the same column and year indicate no significant differences (Tukey's test, $p < 0.05$).

Table 5. Cultivar differences of distribution of dry matter in experiment 1.

Year	Cultivar	Above ground dry matter (g/plant)	Harvest index	Pod dry weight/seed dry weight
2021	Soramizuki	51.5b	0.57	0.28
	Sachiyutaka A1 gou	46.3c	0.56	0.29
	UA4805	57.4a	0.57	0.27
ANOVA		**	ns	ns
2023	Soramizuki	64.4a	0.46	0.31
	Sachiyutaka A1 gou	52.0b	0.47	0.29
	UA4805	56.3ab	0.33	0.62
ANOVA		*	ns	ns
		**	**	ns
		**	ns	ns
Two-way ANOVA	Year	*	ns	ns
	Cultivar	**	ns	ns
	Interaction	*	ns	ns

**: Significantly different ($p < 0.01$). *: Significantly different ($p < 0.05$). ns: Nonsignificant ($p < 0.05$). Same letters within the same column and year indicate no significant differences (Tukey's test, $p < 0.05$).

Table 6. Leaf area and leaf number at the R5 growth stage in experiment 1.

Year	Cultivar	Leaf area (m ² /plant)	Leaf number (/plant)	Leaf area per leaflet (cm ²)
2021	Soramizuki	0.78	58.5a	47.5b
	Sachiyutaka A1 gou	0.84	40.3b	74.3a
	UA4805	0.55	47.7ab	41.0b
ANOVA		ns	**	**
2023	Soramizuki	0.70	66.8a	43.2b
	Sachiyutaka A1 gou	0.56	46.1b	51.5a
	UA4805	0.52	51.2b	40.8b
ANOVA		ns	**	*
		ns	ns	*
		ns	**	**
Two-way ANOVA	Year	ns	ns	*
	Cultivar	ns	**	**
	Interaction	ns	ns	*

**: Significantly different ($p < 0.01$). *: Significantly different ($p < 0.05$). ns: Nonsignificant ($p < 0.05$). Same letters within the same column and year indicate no significant differences (Tukey's test, $p < 0.05$).

no significant differences in leaf area per plant among the cultivars. However, leaf number per plant showed statistically significant differences among the cultivars, with no interaction between year and cultivar. 'Soramizuki' had a larger leaf number compared to 'Sachiyutaka A1 gou' and 'UA4805', while there was no significant difference between 'Sachiyutaka A1 gou' and 'UA4805'. Additionally, 'Soramizuki' and 'UA4805' exhibited significantly smaller leaf sizes (area of one leaflet) than 'Sachiyutaka A1 gou'. 'Sachiyutaka A1 gou', 'Soramizuki' and 'UA4805' are cultivars with three leaflets per leaf.

The results for cultivar differences in PAR within the canopy are shown in Figure 2. Due to differences in canopy height among the cultivars, the top of the canopy was standardized to 100% for each cultivar, and the relationship between relative height and relative PAR was depicted in Figure 2a. In the lower part of the canopy, relative PAR was nearly 0%, with no significant differences among the cultivars (Figure 2a). In contrast, in the upper part of the canopy, namely, in the range from approximately 70% to 100% relative height from the ground (Figure 2a), 'Soramizuki' and 'UA4805' had higher relative PAR than 'Sachiyutaka A1 gou'.

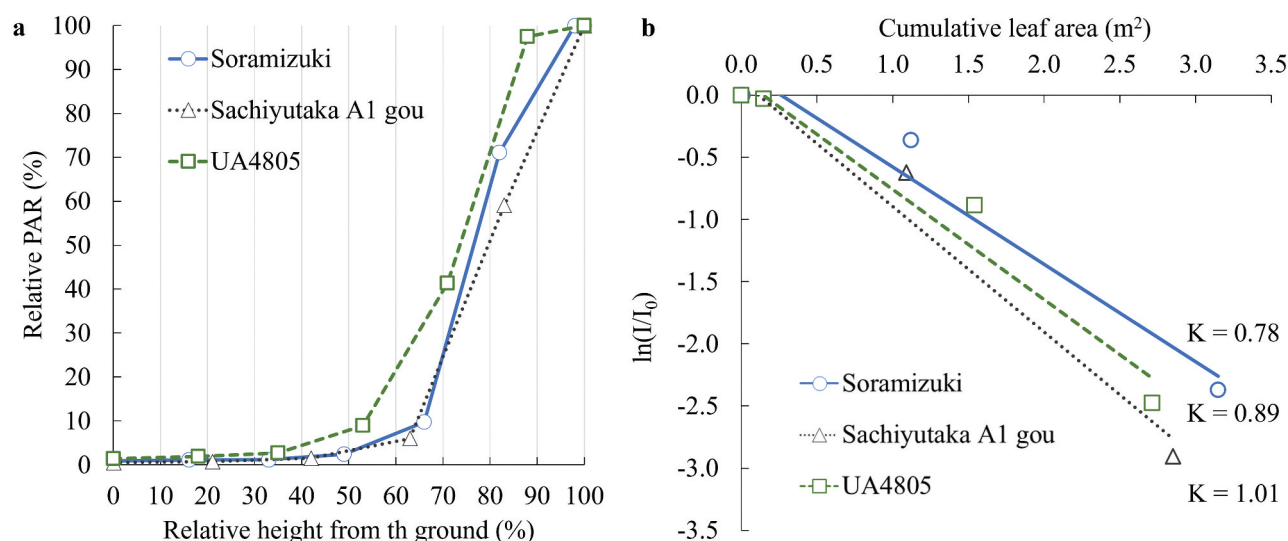


Figure 2. Cultivar difference in relative PAR at the R5 stage. Relative PAR is defined as the ratio of PAR within the canopy to PAR outside the canopy, recorded simultaneously. Relative height from the ground refers to the height expressed as a percentage, with the top of the canopy set at 100% for each cultivar. Cumulative leaf area represents the total leaf area from the ground up to the measurement point. $\ln(I/I_0)$ denotes the natural logarithm of the ratio of PAR inside the canopy to PAR outside the canopy. K is the canopy absorption coefficient calculated as the slope of the regression line for each cultivar in the range from -3.5 to 0, following the method used in a previous study (Shiraiwa et al., 1994).

To compare the ratio of light absorption per leaf area, the relationship between cumulative leaf area and $\ln(I/I_0)$ is shown in Figure 2b. $\ln(I/I_0)$ represents the natural logarithm of the ratio of PAR inside the canopy to PAR outside the canopy (Shiraiwa et al., 1994). The canopy absorption coefficient (K) was calculated as the slope of the regression line for each cultivar in Figure 2b. Significant cultivar differences in relative PAR were only observed in the upper part of the canopy (Figure 2b). Therefore, K was calculated in the range from -3.5 to 0, following the method used in a previous study (Shiraiwa et al., 1994) (Figure 2b). The results showed that K values for 'Soramizuki' (K = 0.78) and 'UA4805' (K = 0.89) were smaller than for 'Sachiyutaka A1 gou' (K = 1.01). This suggests that the light absorption ratio of leaves in 'Soramizuki' and 'UA4805' was lower than in 'Sachiyutaka A1 gou'.

Discussion

Year-to-year differences of yield and yield components in experiment 1

In Experiment 1, the cultivar differences were significant in 2021 but not statistically significant in 2023 (Table 2). This discrepancy was due to the extremely high yield variance of 'UA4805' in 2023 (Table S2). In 2021, seed damage from stink bugs was minimal (0.6%–2.4%) across all cultivars, and its effects on growth parameters were negligible (Table S1). However, in 2023, only

'UA4805' was significantly affected by seed-feeding stink bugs (Table S1). The number of seeds per pod and the weight of 100 seeds, which are directly impacted by seed abortion or damage, significantly decreased in 'UA4805' in 2023 (Table 2). In contrast, the total node number and pod number per node, which are not influenced by seed abortion or damage, showed no year-to-year differences (Table 2).

In 2023, Ibaraki Prefecture reported an above-average number of seed-feeding stink bugs (Control Station for Pests in Ibaraki Prefecture, 2023). Early maturing cultivars tend to suffer more damage from stink bugs (Wada et al., 2006). Therefore, the greater seed damage in 'UA4805' compared to 'Soramizuki' and 'Sachiyutaka A1 gou' was likely due to its earlier maturity. Despite differences in meteorological conditions (Fig. S1) and experimental fields between the years, there was no significant difference in the yields of 'Soramizuki' or 'Sachiyutaka A1 gou'. This suggests that factors other than stink bug damage did not significantly affect the yield and yield components across years or fields in Experiment 1.

Thus, Experiment 1 demonstrated the high yield potential of 'Soramizuki' and 'UA4805' compared to 'Sachiyutaka A1 gou'.

Large sink size of 'soramizuki' and 'UA4805'

Experiment 1 indicated that 'Soramizuki' had a larger sink size than 'Sachiyutaka A1 gou' due to a greater

total node number, pod number per node, and seed number per pod (Table 2). The increased total node number resulted from a rapid rate of node number increase during the vegetative stage, as there were no differences in growth stages between 'Soramizuki' and 'Sachiyutaka A1 gou' (Table 1). The higher pod number per node was suggested to be attributed to a larger flower number per node (Figure 1). The increased seed number was due to a higher proportion of pods with three seed cavities (Table 4).

Both the node numbers on the main stem and branches were larger in 'Soramizuki' and 'UA4805' than in 'Sachiyutaka A1 gou' (Table 2). Node number is controlled by genetic factors (Boerma, 1979). Genetic yield improvements over decades of cultivar development have been correlated with total node number per m² (Kahlon et al., 2011). Kahlon et al. (2011) also reported that reproductive nodes per m², specifically nodes with fertile pods, more accurately distinguish low- and high-yielding cultivars than total node number per m².

Cultivar differences exist in the number of flowers per pod (van Schaik & Probst, 1958). Previous studies have reported cultivar differences in the number of flowers per raceme (Sharma et al., 1990). Dybing (1994) concluded that large and rapidly growing soybean cultivars tend to have a short flowering duration with a high rate of flower production. However, genotypes also possess additional traits that determine flower production characteristics. Experiment 2 suggested that larger pod number per node in 'Soramizuki' and in 'UA4805' than in 'Sachiyutaka A1 gou' derive from cultivar differences in flower number per node (Figure 1), although the further analyses of flower number in branches or in lower node of main stem were necessary to be completely proven.

Yield potential of 'Soramizuki' was not higher than that of 'UA4805', although 'Soramizuki' demonstrated greater total node number per plant and slightly larger 100 seeds weight than 'UA4805' (Table 2). That was because pod number per node of 'Soramizuki' was significantly fewer than that of 'UA4805' (Table 2). The differences in pod number per node between the cultivars exhibited varying trends in Experiment 1 and Experiment 2. In Experiment 1, 'UA4805' had a higher pod number per node compared to 'Soramizuki' (Table 2). Conversely, in Experiment 2, there was no significant difference between 'UA4805' and 'Soramizuki' (Table 3). The data in Experiment 2 was limited to the uppermost three nodes due to time and effort constraints. In contrast, in Experiment 1, the average pod number per node was calculated from the total number of pods on both the main stem and branches (Table 2). Flower buds per node tend to be more numerous at the upper nodes of the main stem

compared to the nodes on branches (Saitoh et al., 1998). This suggests that 'Soramizuki' may produce fewer flowers or pods per node on its branches compared to 'UA4805'. Further studies are necessary to analyze the pod number and the flower number at nodes of branches in 'Soramizuki' to identify ways to improve its yield potential.

There were no significant differences in the seed number per pod cavity among the three cultivars (Table 4). Therefore, the higher ratio of pods with three pod cavities was the cause of the larger seed number per pod in 'Soramizuki' and 'UA4805' than in 'Sachiyutaka A1 gou' (Table 2). The fertile seed ratio can be influenced by environmental factors, such as water stress (Doss et al., 1974; Sionit & Kramer, 1977) or temperature stress after the R1 stage (Gibson & Mullen, 1996; Mochizuki et al., 2005). In contrast, the ratio of pods with three pod cavities is known to be genetically regulated (Chanchu et al., 2023; Jeong et al., 2012; Sayama et al., 2017). The gene *Ln* controls both leaflet shape and seed number per pod (Jeong et al., 2012). Loss of function in this gene (*ln/ln*) results in longer leaflets and a higher seed number per pod compared to the wild type (Sayama et al., 2017). Chanchu et al. (2023) reported that *Ln* decreases the seed number per pod due to a trade-off between the number of pods with two pod cavities and those with four pod cavities. It is speculated that the genetic factor for a larger seed number per pod in 'UA4805' was inherited by 'Soramizuki'. However, 'Soramizuki' and 'Sachiyutaka A1 gou' share the same leaf shape category, 'pointed ovate' (Kato et al., 2023), suggesting that *Ln* is not related to the differences among the three cultivars. Further studies comparing the genotype of *Ln* and detailed leaf shape (ratio of length to width) are needed.

The results of yield components (node number, pod number per node, seed number per pod) suggest that the large sink size of 'Soramizuki' and 'UA4805' is genetically regulated, indicating that the large sink size was genetically inherited from 'UA4805' to 'Soramizuki'. However, the discussion is limited as only phenotype data were used. Further analysis of the relationships between phenotypes and genotypes for each factor is expected. Kato et al. (2015) indicated that indeterminate phenotype caused by *Dt1* gene is a useful for breeding high-yielding cultivars because of improved seed number per plant and pod number per plant, although indeterminate phenotype was susceptible to lodging because of its tall plant height due to elongation of the main stem after flowering. On the other hand, 'UA4805' was determinate phenotype. Thus, the present study indicated that genetic factors in US cultivar other than *Dt1* could be effective for breeding high-yielding cultivars in Japan.

Source ability of 'Soramizuki' and 'UA4805'

Soybean yields are determined by total dry matter accumulation and the partitioning of assimilates to the seeds, as measured by the harvest index (Kumudini, 2002). The contribution of dry matter accumulation and harvest index to soybean yield varies across studies (Kumudini, 2002). Morrison et al. (1999) found that increased soybean yield in Canada was significantly correlated with an improved harvest index. Jin et al. (2010) reported that yield improvement in Northeast China was associated with increases in both plant dry weight and harvest index. De Bruin and Pederson (2009) indicated that enhanced total dry matter accumulation led to yield improvement in Iowa. Kawasaki et al. (2016) compared Japanese and US soybean cultivars and found that yield differences were more closely related to total dry matter accumulation than to the harvest index.

In the present study, no cultivar differences in harvest index were observed (Table 5). Kawasaki et al. (2018) compared 'UA4805' and 'Sachiyutaka' and found that 'UA4805' did not exhibit a higher harvest index. The high total dry matter accumulation of 'UA4805' is suggested to have been inherited by 'Soramizuki.'

Single-leaf photosynthetic rates can vary between cultivars (Buttery et al., 1981). Breeding efforts have improved these rates (Jin et al., 2010; Morrison et al., 1999). Tanaka and Shiraiwa (2009) exhibited that indeterminate phenotype, caused by *Dt1* gene, resulted in a greater stomatal conductance than determinate phenotype. Tanaka et al. (2010) indicated that stomatal conductance, which correlates with stomatal density on a leaf, varied among cultivars and was related to photosynthetic rates. In a study comparing the photosynthetic rates of Japanese and US cultivars, 'UA4805' exhibited the highest photosynthetic rate, attributed to high stomatal conductance and increased photosynthetic efficiency under fluctuating light conditions (Soleh et al., 2016). In the future study, genes related to the single-leaf photosynthetic rate and stomatal conductance of 'Soramizuki' should be verified because genes other than *Dt1* may be identified as the useful for high-yielding.

Light interception and distribution within a plant community are critical for crop productivity. Many lower soybean leaves do not receive adequate radiation due to self-shading, where the upper canopy intercepts light (Sakamoto & Shaw, 1967). The inclination angle of leaflets, influenced by active leaf diurnal movement, was suggested to be important for light penetration into the lower canopy layers (Saitoh et al., 1995). Smaller leaflets in the upper canopy of determinate cultivars may also enhance radiation use efficiency (Isoda et al., 1994). In this study, cultivar differences of leaf area per plant were

not statistically significant (Table 6) and its tendency varied depending on year. For example, 'Sachiyutaka' showed larger leaf area per plant than 'Soramizuki' in 2021 but it showed smaller leaf area per plant than 'Soramizuki' in 2023. Thus, the relationship between leaf area per plant and yield was unclear in the present study. On the other hand, it was found that 'UA4805' and 'Soramizuki' had more leaves and smaller leaflets than 'Sachiyutaka A1 gou,' suggesting an efficient leaf distribution pattern in the canopies of 'Soramizuki' and 'UA4805.' It was suggested that 'Soramizuki' and 'UA4805' had better light penetration in the upper canopy compared to 'Sachiyutaka A1 gou' (Figure 2). Therefore, it is suggested that 'Soramizuki' inherited superior light interception characteristics from 'UA4805,' although further analysis of leaf inclination, size, and distribution in 'Soramizuki' is needed.

Conclusions

The present study indicated that the high-yielding factors of 'Soramizuki' and 'UA4805' are attributed to their larger sink size and higher source ability compared to 'Sachiyutaka A1 gou'. This suggests that breeding efforts to incorporate high-yielding traits from US cultivars into Japanese cultivars have been successful. The high-yielding traits were related to genes other than *Dt1* genes because 'UA4805' was a determinate cultivar. However, 'Soramizuki' exhibited lower above-ground dry matter and yield potential than 'UA4805', despite having a later maturity than 'UA4805' (Tables 1, 2, 5). This implies that 'Soramizuki' did not fully inherit the high-yielding capacity of 'UA4805'. 'Soramizuki' also had a lower pod number despite a greater total node number compared to 'UA4805'. Further research is needed to investigate the cause of the reduced pod number in 'Soramizuki.' This study focused on phenotypic differences among cultivars; therefore, future research should aim to correlate these phenotypes with specific genes. The development of subsequent cultivars derived from US varieties with even higher yield potential is anticipated. This study provides valuable insights for effective breeding selection.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Ryoma Takeshima  <http://orcid.org/0000-0002-2525-7334>

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